gpeyt1-24 · Geotextile wall spacing

GEO PE · Long Calculation Problems I — Topic 5: Earth Structures, Ground Improvement & Pavements

Problem

A 16 ft geotextile-wrapped reinforced wall retains granular backfill (γ = 120 pcf, φ = 32°). A uniform surcharge q = 250 psf acts on top (no live point loads). The geotextile has wide-width Tult = 2,400 lb/ft and cumulative reduction factors ΠRF = 2.6. Use FS = 1.5 on reinforcement. Check the bottom layer at Z = 16 ft. Facing batter is vertical; foundation is dense gravel.

Most nearly, the maximum vertical spacing of the bottom layer is:

A. 2.40 ft
B. 1.04 ft
C. 0.09 ft
D. 0.92 ft

Given equations / data

```
σh = Ka γ Z + Ka q; Tallow = Tult/ΠRF; Sv = Tallow/(σh · FS); Ka = tan²(45 − φ/2)
```

Worked solution

1. Ka = tan²(45 − 32/2) = 0.307
2. σh = 0.307×120×16 + 0.307×250 = 667 psf
3. Tallow = 2,400/2.6 = 923 lb/ft
4. Sv = 923/(667×1.5) = 0.92 ft (≈ 11 in.)

Final answer

0.92 ft

Why the other values are wrong

- **2.40 ft** — used Tult without reduction factors
- **1.04 ft** — omitted the surcharge pressure
- **0.09 ft** — used Kp in place of Ka

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.13.5.1 Geotextile-Reinforced Walls, p. 170

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=4cCtjNpcFkA (no creator values copied).

Personal use only — draft study material for Jayce Wright. Not for redistribution. Not reviewed by a licensed PE.